

S0

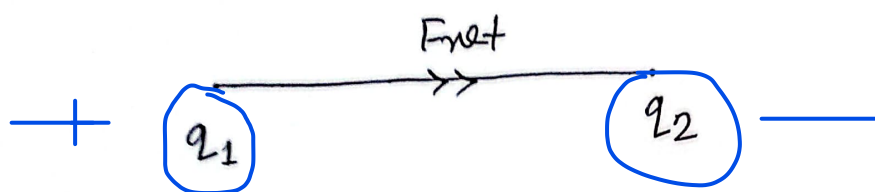
①

① Net electric force at q_1 is 5.7 N

Now,

$$5.7 = 9 \times 10^9 \times \frac{26 \times 10^{-6} \times 47 \times 10^{-6}}{r^2}$$

$$\therefore r = 1.4 \text{ m}$$



It is attractive force.

②

$$F_1 = F_2$$

(a) $m_1 a_1 = m_2 a_2$

$$\Rightarrow 6.3 \times 10^{-7} \times 7 = m_2 \times 9$$

$$\therefore m_2 = 4.9 \times 10^{-7} \text{ kg}$$

(b) $6.3 \times 10^{-7} \times 7 = 9 \times 10^9 \frac{q^2}{r^2}$

$$\therefore q = 7.1 \times 10^{-11} \text{ C} \quad [r = 3.2 \times 10^{-3} \text{ m}]$$

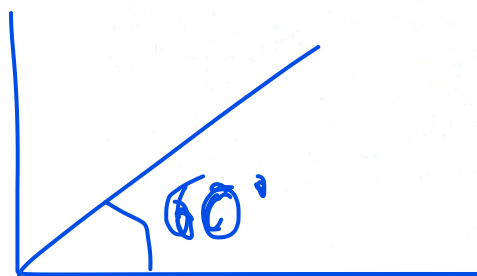
②

③

$$(i) F_x = F \cos 60 = 40$$

$$\therefore F = 80 \text{ N}$$

$$\therefore F_y = F \sin 60 = 80 \times \frac{\sqrt{3}}{2} = 40\sqrt{3} \text{ N}$$



(ii)

$$\vec{F}_1 \cdot \vec{F}_2 = |\vec{F}_1| |\vec{F}_2| \cos \theta$$

$$\Rightarrow (3\hat{i} - 4\hat{j}) \cdot (-2\hat{i} + 3\hat{k}) = \frac{(\sqrt{3^2 + (-4)^2})(\sqrt{(-2)^2 + 3^2})}{\cos \theta}$$

$$\Rightarrow -6 = (5\sqrt{13}) \cos \theta$$

$$\therefore \theta = \cos^{-1} \left\{ -6 / (5\sqrt{13}) \right\}$$

$$= 153.03^\circ \text{ or } 109.44^\circ$$

Normal

(3)

$$\textcircled{4} \quad F_x = -25 \text{ m}, \quad F_y = 40 \text{ m}$$

$$(a) \quad F = \sqrt{F_x^2 + F_y^2} = \sqrt{(-25)^2 + 40^2} = \underline{47.17 \text{ m}}$$

$$(b) \quad \theta = \tan^{-1}(40/-25) = -57.99^\circ \approx -58^\circ \quad \checkmark$$

$$\textcircled{5} \quad F_1 = 5 \text{ N}, \quad \alpha = 120^\circ, \quad \theta = 90^\circ \text{ (with } F_2 \text{)}$$

$$\therefore \tan 90 = \frac{F_1 \sin 120^\circ}{F_2 + F_1 \cos 120}$$

$$\Rightarrow \frac{1}{0} = \frac{5 \sin 120}{F_2 + F_1 \cos 120}$$

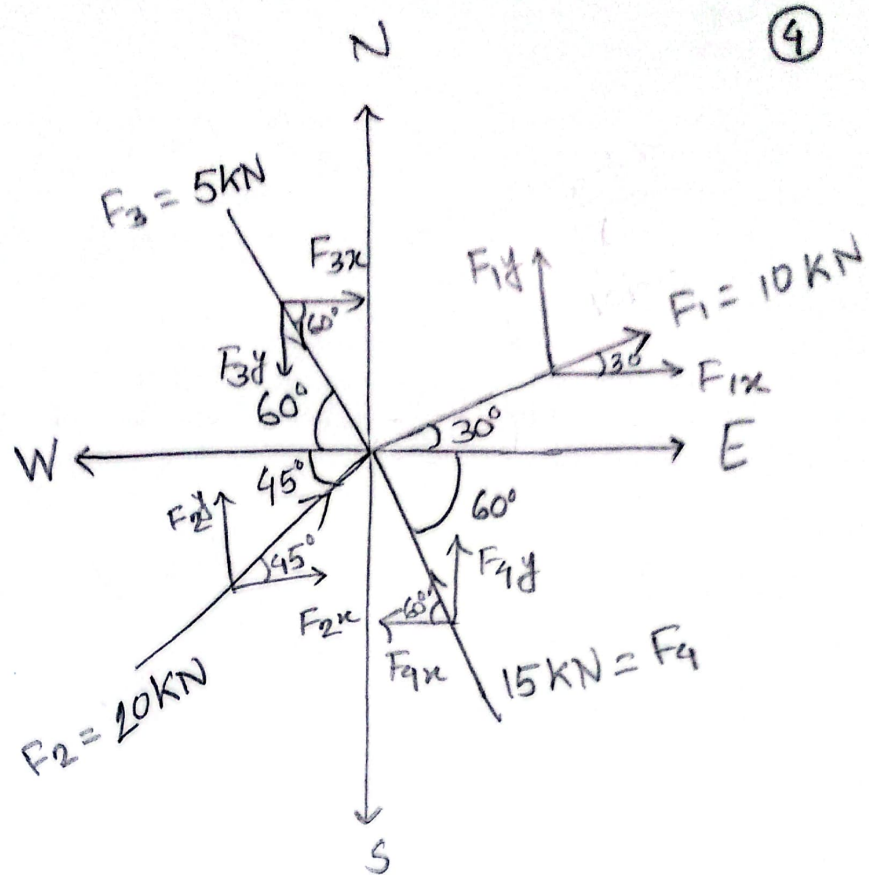
$$\therefore F_2 = 2.5 \text{ N}$$

$$\therefore F_{\text{net}} = \sqrt{5^2 + (2.5)^2 + 2 \cdot 5 \cdot (2.5) \cos 120}$$

$$= 5\sqrt{3}/2$$

$$= 4.33 \text{ N}$$

⑥



$$\begin{aligned} \therefore F_{x \text{ net}} &= 10 \cos 30 + 20 \cos 45 + 5 \cos 60 \\ &\quad - 15 \cos 60 \\ &= 17.8 \text{ kN} \end{aligned}$$

$$\begin{aligned} F_{y \text{ net}} &= 10 \sin 30 + 20 \sin 45 - 5 \sin 60 + 15 \sin 60 \\ &= 27.8 \text{ kN} \end{aligned}$$

$$\therefore F_{\text{net}} = \sqrt{17.8^2 + 27.8^2} = 33.01 \text{ kN}$$

$$\theta = \tan^{-1} \left(\frac{27.8}{17.8} \right) = 57.37^\circ$$

(5)

⑦ Let, $q_1 = q$, $q_2 = 2q$,

initial distance = x

$$\therefore 3 \times 10^{-3} = k \frac{2q^2}{x^2} \quad \text{--- (i)}$$

$$\therefore 5 \times 10^{-4} = k \frac{2q^2}{(x+0.1)^2} \quad \text{--- (ii)}$$

(i) $\div$ (ii)

$$\frac{3 \times 10^{-3}}{5 \times 10^{-4}} = \frac{k \frac{2q^2}{x^2}}{k \frac{2q^2}{(x+0.1)^2}}$$

$$\Rightarrow 6 = \frac{(x+0.1)^2}{x^2}$$

$$\Rightarrow 5x^2 - 0.2x - 0.01 = 0$$

$$\Rightarrow x = 0.07, -0.03$$

$\therefore x = 0.07 \text{ m}$ [$\because$ distance cannot be negative]

$$x = 0.07 \rightarrow \text{(i)}$$

$$q = 2.86 \times 10^{-8} \text{ C}$$

$$\therefore q_1 = 2.86 \times 10^{-8} \text{ C}$$

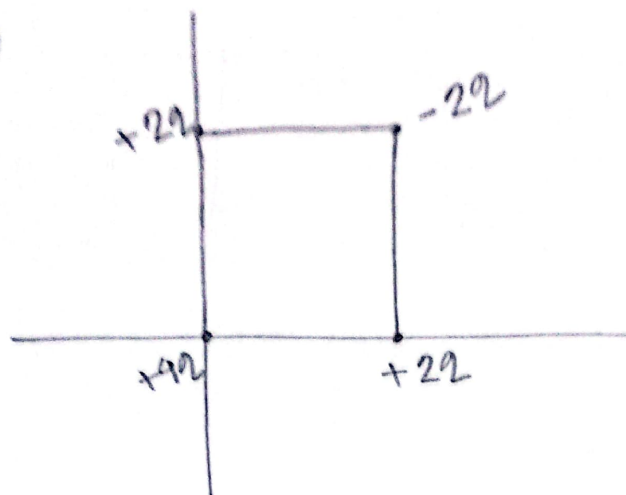
$$q_2 = 5.7 \times 10^{-8} \text{ C}$$

⑥

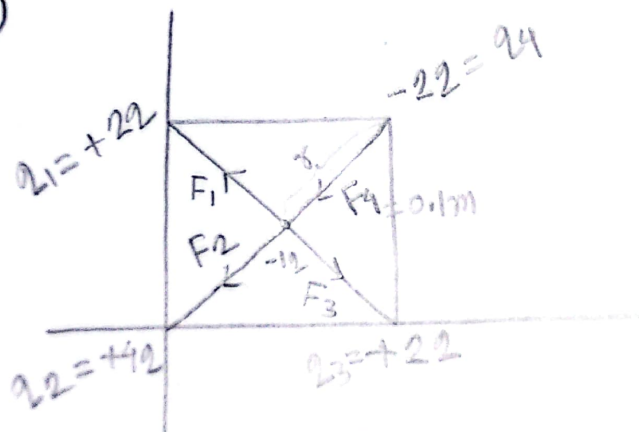
⑧ Given,

$$q_1 = +2q, q_2 = +4q, q_3 = +2q, q_4 = -2q$$

(i)



(ii)



here,

$$r = \frac{0.1}{\sqrt{2}} = 0.07m$$

Here, F_1 and F_3 cancel out each others.

$$F_2 = k \frac{4q \times q}{(0.07)^2}, F_4 = k \frac{2q \times q}{(0.07)^2}$$

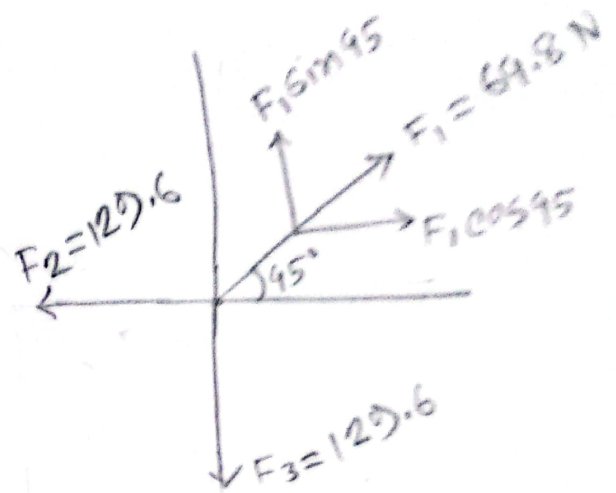
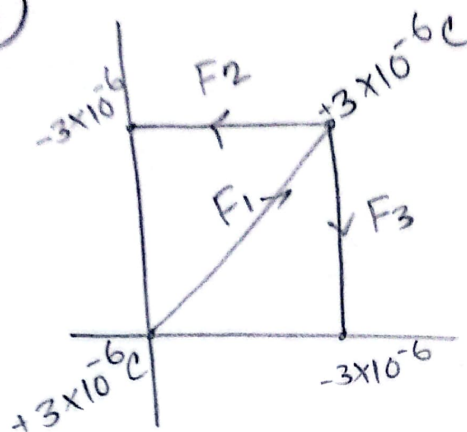
$$\therefore F_{net} = F_2 + F_4 \quad [\because F_2 \text{ and } F_4 \text{ same direction}]$$

$$= \frac{k \times 4q \times q}{(0.07)^2} + \frac{k \times 2q \times q}{(0.07)^2}$$

$$= k \frac{(3 \times 10^{-9})^2}{(0.07)^2} (4+2) = 9.9 \times 10^{-5} N$$

⑦
according to the diagram, ~~net ele~~ The direction of electrostatic force work at -22 to $+42$.

⑧



$$F_1 = 9 \times 10^9 \frac{(3 \times 10^{-6})^2}{(\sqrt{2} \times 2.5 \times 10^{-2})^2}$$

$$= 69.8 \text{ N}$$

$$F_2 = F_3 = \frac{9 \times 10^9 \times (3 \times 10^{-6})^2}{(2.5 \times 10^{-2})^2} = 120.6 \text{ N}$$

$$\therefore F_{x\text{net}} = 69.8 \cos 45 - 120.6 = -83.78$$

$$F_{y\text{net}} = 69.8 \sin 45 - 120.6 = -83.78$$

$$\therefore F_{\text{net}} = \sqrt{(-83.78)^2 + (-83.78)^2} = 118.48 \text{ N}$$

$$\therefore \theta = \tan^{-1} \left(\frac{-83.78}{-83.78} \right) = 225^\circ$$

(8)

(10)

(i) The charge of $(1.66 \times 10^{-24} \times 23) \text{ gm Na}$ is $(11 \times 1.6 \times 10^{-19}) \text{ C}$

So, The charge of 4 gm Na is $\frac{4 \times 11 \times 1.6 \times 10^{-19}}{1.66 \times 10^{-24} \times 23} \text{ C}$
 $= 1.84 \times 10^5 \text{ C}$

$$\therefore F = 9 \times 10^9 \frac{(1.84 \times 10^5)^2}{2^2} = 7.6 \times 10^{19} \text{ N}$$

(ii)

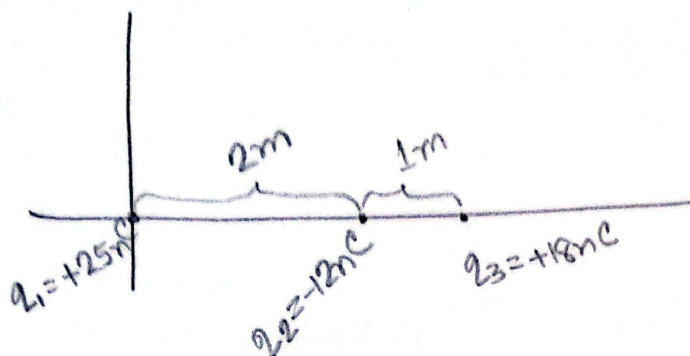
$$F = 7.6 \times 10^{19}$$

$$\Rightarrow ma = 7.6 \times 10^{19}$$

$$\Rightarrow a = \frac{7.6 \times 10^{19}}{4} = 1.9 \times 10^{19} \text{ m/s}^2$$

(9)

(11)



$$F_1 = k \frac{q_1 q_2}{2^2}$$

$$F_2 = k \frac{q_1 q_3}{3^2}$$

$$\therefore F = F_2 - F_1$$

$$= k \frac{q_1 q_3}{3^2} - k \frac{q_1 q_2}{2^2}$$

$$= 9 \times 10^9 \times 25 \times 10^{-9} \left\{ \frac{18}{3^2} - \frac{12}{2^2} \right\}$$

$$= -2.25 \times 10^{-7} \text{ N}$$

$$\therefore |F| = 2.25 \times 10^{-7} \text{ N}$$

The direction at +ve x-axis.

(10)

$$(12) \quad F_E = 9 \times 10^9 \frac{(1.6 \times 10^{-19})^2}{(5 \times 10^{-9})^2} = 0.2 \times 10^{-12} \text{ N}$$

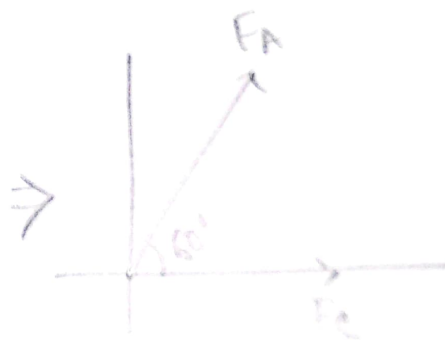
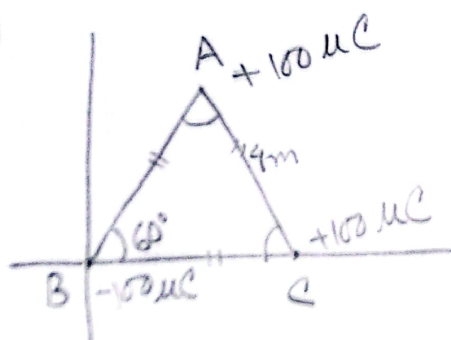
$$F_G = 6.6743 \times 10^{-11} \left\{ \frac{1.67262192 \times 10^{-27}}{5 \times 10^{-9}} \right\}^2$$

$$= 7.5 \times 10^{-48} \text{ N}$$

$$\frac{F_E}{F_G} = \frac{0.2 \times 10^{-12}}{7.5 \times 10^{-48}} = 1.227 \times 10^{36}$$

$$\therefore F_E : F_G = 1.227 \times 10^{36} : 1$$

(13)



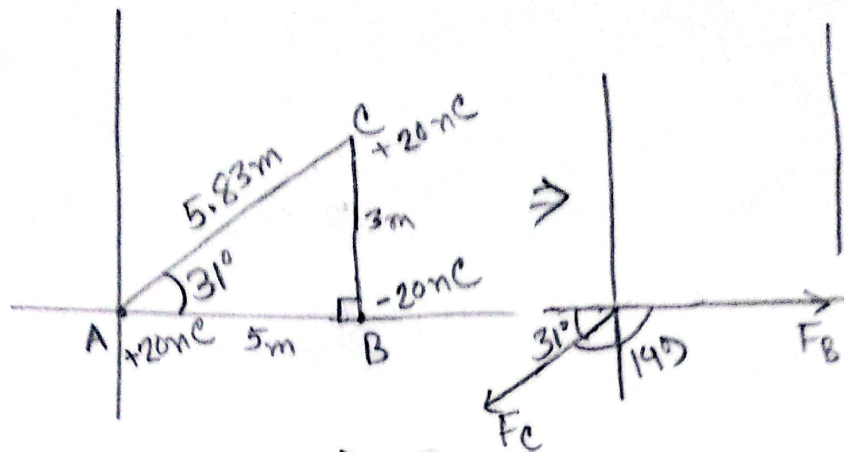
$$F_C = F_A = 9 \times 10^9 \left\{ \frac{100 \times 10^{-6}}{4} \right\}^2 = 5.625 \text{ N}$$

$$\therefore F_{\text{net}} = \sqrt{2 \times (5.625)^2 \times (1 + \cos 60)} = 9.74 \text{ N}$$

$$\theta = \tan^{-1} \left\{ \frac{5.625 \sin 60}{5.625 + 5.625 \cos 60} \right\} = 30^\circ \text{ [with +ve x-axis]}$$

(11)

(14)



$$\cos \theta = \frac{5}{5.83}$$

$$\therefore \theta = 30.96376$$

$$\approx 31^\circ$$

$$F_c = 9 \times 10^9 \left\{ \frac{20 \times 10^{-9}}{5.83} \right\}^2 = 1.06 \times 10^{-7} \text{ N}$$

$$F_b = 9 \times 10^9 \left\{ \frac{20 \times 10^{-9}}{5} \right\}^2 = 1.44 \times 10^{-7} \text{ N}$$

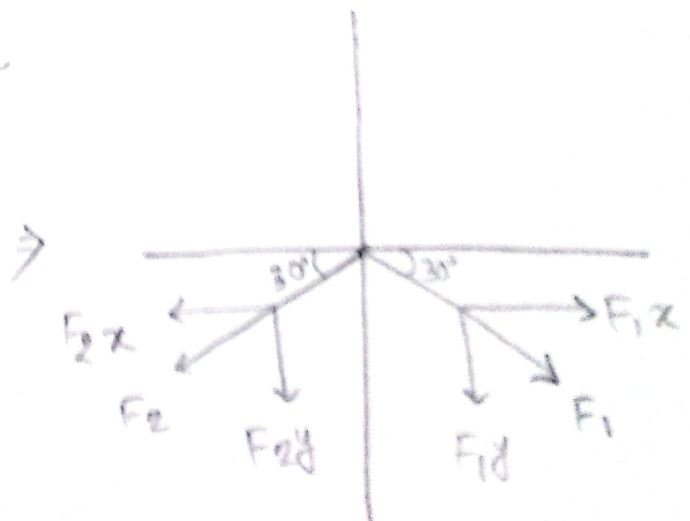
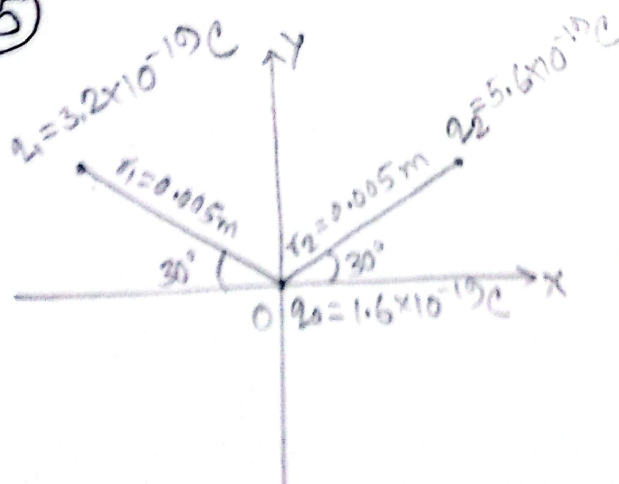
$$F_{\text{net}} = \sqrt{F_c^2 + F_b^2 + 2F_b F_c \cos(149^\circ)}$$

$$= 7.62 \times 10^{-8} \text{ N}$$

$$\theta = \tan^{-1} \left\{ \frac{1.06 \times 10^{-7} \sin(149^\circ)}{1.44 \times 10^{-7} + 1.06 \times 10^{-7} \cos(149^\circ)} \right\}$$

$$= 45.77^\circ \text{ [clockwise with +ve x-axis]}$$

(15)



(12)

$$F_1 = 9 \times 10^9 \frac{3.2 \times 10^{-19} \times 1.6 \times 10^{-19}}{(0.005)^2} = 1.8432 \times 10^{-23} \text{ N}$$

$$F_2 = 9 \times 10^9 \frac{5.6 \times 10^{-19} \times 1.6 \times 10^{-19}}{(0.005)^2} = 3.2256 \times 10^{-23} \text{ N}$$

$$F_{x\text{net}} = F_{1x} - F_{2x}$$

$$= 1.8432 \times 10^{-23} \cos 30 - 3.2256 \times 10^{-23} \cos 30$$

$$= -1.107 \times 10^{-23} \text{ N}$$

$$F_{y\text{net}} = -1.8432 \times 10^{-23} \sin 30 - 3.2256 \times 10^{-23} \sin 30$$

$$= -2.5344 \times 10^{-23} \text{ N}$$

$$\therefore F_{\text{net}} = \sqrt{(F_{x\text{net}})^2 + (F_{y\text{net}})^2} = 2.8 \times 10^{-23} \text{ N}$$

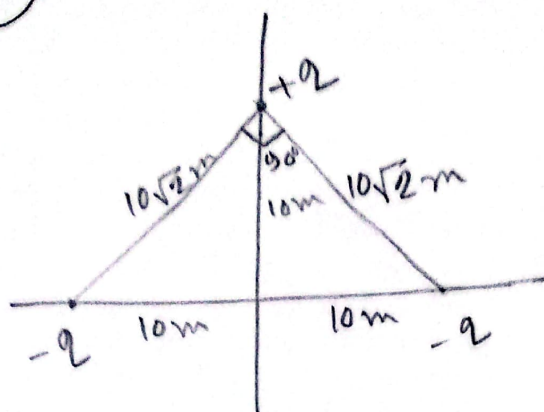
$$\therefore \theta = \tan^{-1} \left\{ \frac{F_{y\text{net}}}{F_{x\text{net}}} \right\} = 180 + 64.72 = 244.72^\circ$$

(16)

$$F = 9 \times 10^9 \left\{ \frac{1.6 \times 10^{-19}}{3.81 \times 10^{-7}} \right\}^2$$

$$= 1.6 \times 10^{-15} \text{ N}$$

17



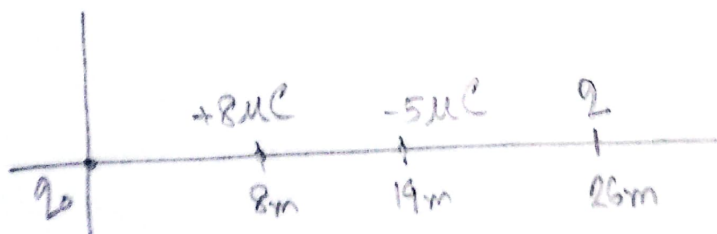
$$q = 10 \times 10^{-5} \text{ C}$$

$$(i) F_1 = F_2 = 9 \times 10^9 \left\{ \frac{10 \times 10^{-5}}{10\sqrt{2}} \right\}^2 = 0.45 \text{ N}$$

$$\therefore F = \sqrt{F_1^2 + F_2^2} = 0.6364 \text{ N}$$

(ii) Direction of force at downward along the y-axis.

18



Let, test charge = q_0
 $\&$ q charge placed at $x = 26 \text{ m}$.

Now,

$$F_{(8)} - F_{(-5)} = -F_{(q)}$$

$$\Rightarrow k \frac{8 \times 10^{-6} \times q_0}{8^2} - k \frac{5 \times 10^{-6} \times q_0}{14^2} = -k \frac{q q_0}{26^2}$$

$$\Rightarrow \left(\frac{8}{8^2} - \frac{5}{14^2} \right) \times 10^{-6} = - \frac{q}{26^2}$$

$$\therefore q = -67.255 \times 10^{-6} \text{ C}$$

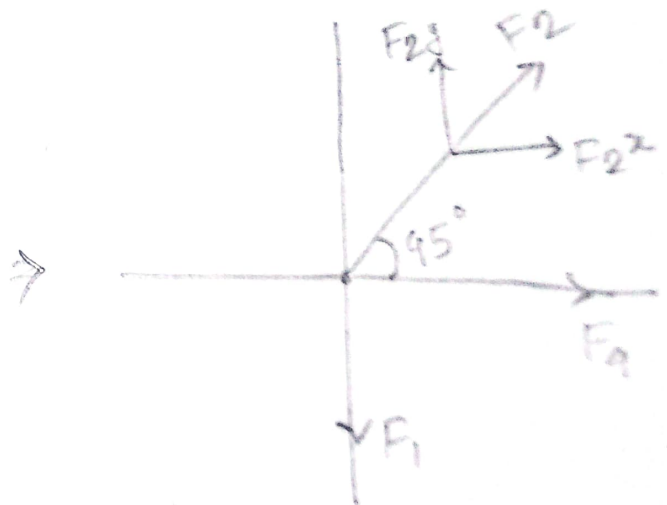
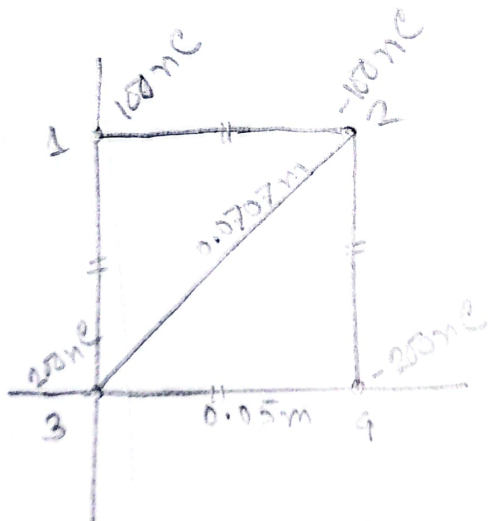
$$\therefore q = -67.255 \mu\text{C}$$

(19)

$$\frac{F_E}{F_G} = \frac{(1.6 \times 10^{-19})^2}{r^2} = 0.8 \times 1.6726 \times 10^{-27}$$

$$\therefore r = 0.1186 \text{ m}$$

(20)



$$F_1 = 9 \times 10^9 \frac{20000 \times 10^{-9}}{(0.05)^2} = 7.2 \times 10^{-7} \text{ N } 0.072 \text{ N}$$

$$F_2 = 9 \times 10^9 \frac{2 \times 10^{-14}}{(0.0707)^2} = 0.036 \text{ N}$$

$$F_4 = 9 \times 10^9 \frac{4 \times 10^{-14}}{(0.05)^2} = 0.144 \text{ N}$$

$$(a) F_{xnet} = F_{2x} + F_4$$

$$= F_2 \cos 45^\circ + F_4$$

$$= 0.1695 \text{ N}$$

$$(b) F_{ynet} = F_{2y} - F_1$$

$$= F_2 \sin 45^\circ - F_1$$

$$= -0.0465 \text{ N}$$